MIL-G-50349A(MU)
26 April 1976
SUPERSEDING
MIL-G-50349(MU)
10 November 1969

MILITARY SPECIFICATION

GENERATOR, PULSE: 10554892

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a pulse generator, hereinafter called gate pulse generator, used in Test Set, Subassembly MX-8048/TPM-22, part of Radar Set AN/TPM-22.

2. APPLICABLE DOCUMENTS

2.2 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel: Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction: M18 Manufacturing Process and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U. S. Army, Frankford Arsenal

FSC 4931

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F10554527 Test Set, Subassembly MX-8048/TPM-22

F10554892 Gate Pulse Generator

PACKAGING DATA SHEET

10554892 Packaging of Gate Pulse Generator

(Copies of specification, standards, drawings and packaging data sheets required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The gate pulse generator shall be manufactured in accordance with Drawing F10554892 and drawings pertaining thereto.

3.2 General specification.- Where applicable, the gate pulse generator shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishings
d. Part identification and marking
e. Workmanship
f. Soldering electrical connections

3.2.2 MIL-C-46357

a. Etched circuit assemblies
b. Wiring
c. Threaded fasteners
d. Marking
e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration.- The gate pulse generator shall be mounted on the vibration table in normal service position. The gate pulse generator shall meet the requirements of 3.4 and 3.5 after being subject to continuous vibration for two hours along each of three mutually perpendicular axes within the frequency range and amplitude as follows:

Frequency Range	Double Amplitude
5 to 15	0.06 inch
15 to 25	0.04 inch
25 to 55	0.02 inch

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The rate of change of frequency at each frequency range shall not exceed one cycle-per-second (1 cps). Vibration tests shall be conducted under ambient room conditions.

3.3.2 Shock.- The gate pulse generator shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G's) and a pulse duration of 11 ± 1 milliseconds.

3.3.3 Altitude.

3.3.3.1 Operating.- The gate pulse generator shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Non-operating.- The gate pulse generator shall meet the requirements of 3.4 and 3.5 after exposure for not less than 1 hour to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating.- The gate pulse generator shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between +32°F and +131°F.

3.3.4.2 Non-operating (storage).- The gate pulse generator shall not suffer permanent damage when subjected to the temperature extremes of -80°F and +185°F, and shall meet the requirements of 3.4 and 3.5 when turned to room ambient.

3.3.5 Humidity-non-operating.- The gate pulse generator shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical.- Each conductor shall be electrically continuous as specified on the schematic diagram.

3.5 Performance.- When the gate pulse generator is supplied with the input voltages of Table I, application of the input signals of Table I to the gate pulse generator shall cause it to yield the output signals indicated in Table I.

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TABLE I - INPUT AND OUTPUT CHARACTERISTICS

REQUIREMENT	CHARACTERISTICS	TERMINALS
Input voltages		
+5V	5Vdc ±0.300	P1-30,31,32(+) P1-24,25,26(RET)
+25V	25Vdc ±0.300	P1-55,56(+): P1-53,54(RET)
Input signal		
Cp	7.5MHz±1% square wave or sine wave, amplitude 4.3-5.2 volts peak-to-peak	P1-43: P1-44(RET)
Output signals in response to Cp		
Cp out	Same as Cp	P1-47: P1-22(RET)
Cpd out	7.5MHz±1% square wave, amplitude 4.3-5.2 volts peak-to-peak, delayed 20-50 ns from Cp, with 150 ohm load	P1-45: P1-22(RET)
Input signal PRF	20KHz±1% negative pulse (from not less than +3Vdc level to 0V and return), approx 0.6 usec wide	P1-41,42: P1-44(RET)
Output signal in response to PRF	Pulse train 20KHz±1%, pulse width not less than 55 ns and not more than 85 ns not less than +12V amplitude, 60 ns max rise time, with 150 ohm load	P1-37,39: P1-52(RET)
Input signal TARGET PULSE	Same as PRF	P1-14,15: P1-44(RET)
Output signal in response to TARGET PULSE	Pulse train 20KHz±1%, pulse width not less than 55 ns and not more than 85 ns, not less than +12V amplitude, not more than 150 ns rise time, with 150 ohm load	P1-17,18: P1-22(RET)

NOTE: Input frequencies need not be exactly those given above. However, output frequency of given signal should be same as frequency of its corresponding input signal.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of one gate pulse generator manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot information, and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

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4.3.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables II and III herein. Examinations and test for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE II - CLASSIFICATION OF DEFECTS

100% INSPECTION

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101. Electrical	3.4	4.6.2
102. Performance	3.5	4.6.3

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.- One gate pulse generator out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests in Table II and shall then meet the requirements and tests in Table III.

TABLE II - SPECIAL SAMPLING

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Fabrication	3.1	Applicable dwg. visual
302. General specifications	3.2	MIL-F-13926/ MIL-C-46357
303. Vibration	3.3.1	4.6.1
304. Shock	3.3.2	4.6.1.2
305. Altitude - operating	3.3.3.1	4.6.1.3
306. Altitude - nonoperating	3.3.3.2	4.6.1.3

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307. Temperature	3.3.4	4.6.1.4
308 Humidity - non-operating	3.3.5	4.6.1.5

4.4.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government-furnished inspection equipment.- Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor-furnished inspection equipment.

4.5.2.1 Contractor design.- The contractor shall design and supply inspection equipment compatible with "Test Methods an Procedures" specified in 4.6 of this specification and with the "Component Inspection" procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is considered normally to be within 10 percent of product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to ensure that the test equipment will indicate reliable acceptability of a product that does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of the test equipment.

4.5.2.2 Test equipment.- In conjunction with 4.5.2.1, the following test equipment shall be used in the performance of the applicable test as specified in 4.6:

TEST EQUIPMENT LIST

Item No.	Description
1. Power supply	5.0 ± 0.3 Vdc, regulated; ac ripple less than 50 mV peak-to-peak: capacity 300 mA.
2. Power supply	25.0 ± 0.3 Vdc, regulated; ac ripple less than 100 mV peak-to-peak; capacity 500 mA.

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Item No.	Description
3. Signal generator	7.5 MHz square wave or sine wave, amplitude 5.0 ± 0.2 V peak-to-peak, into 50 ohm load.
4. Oscilloscope	Dual channel, vertical passband DC to 15 MHz. Rise time of scope plus measuring probe 10 nanoseconds or less. Sweep period 50 nanoseconds per centimeter or less. Input resistance greater than 8 megohms. Input capacitance less than 25 picofarads.
5. Pulse generator	20 KHz pulse train. Negative going pulses from +3 Vdc level to 0 V and return, into 150 ohms, approx. 0.6 usec wide.
6. Test harness	Per Figure 1.

PARTS LIST FOR TEST HARNESS, ITEM 6

Desig.	Part	Description
P1	Connector	Malco 5803232-1 Elco 07-5422-0198-01, or equivalent
S1	Switch	SPDT
S2	Switch	Rotary, three-position
R1	Resistor	150 ohms, 3 W
R2	Resistor	150 ohms, 3 W
R3	Resistor	150 ohms, 3 W

Wire, mounting hardware, per Figure 1.

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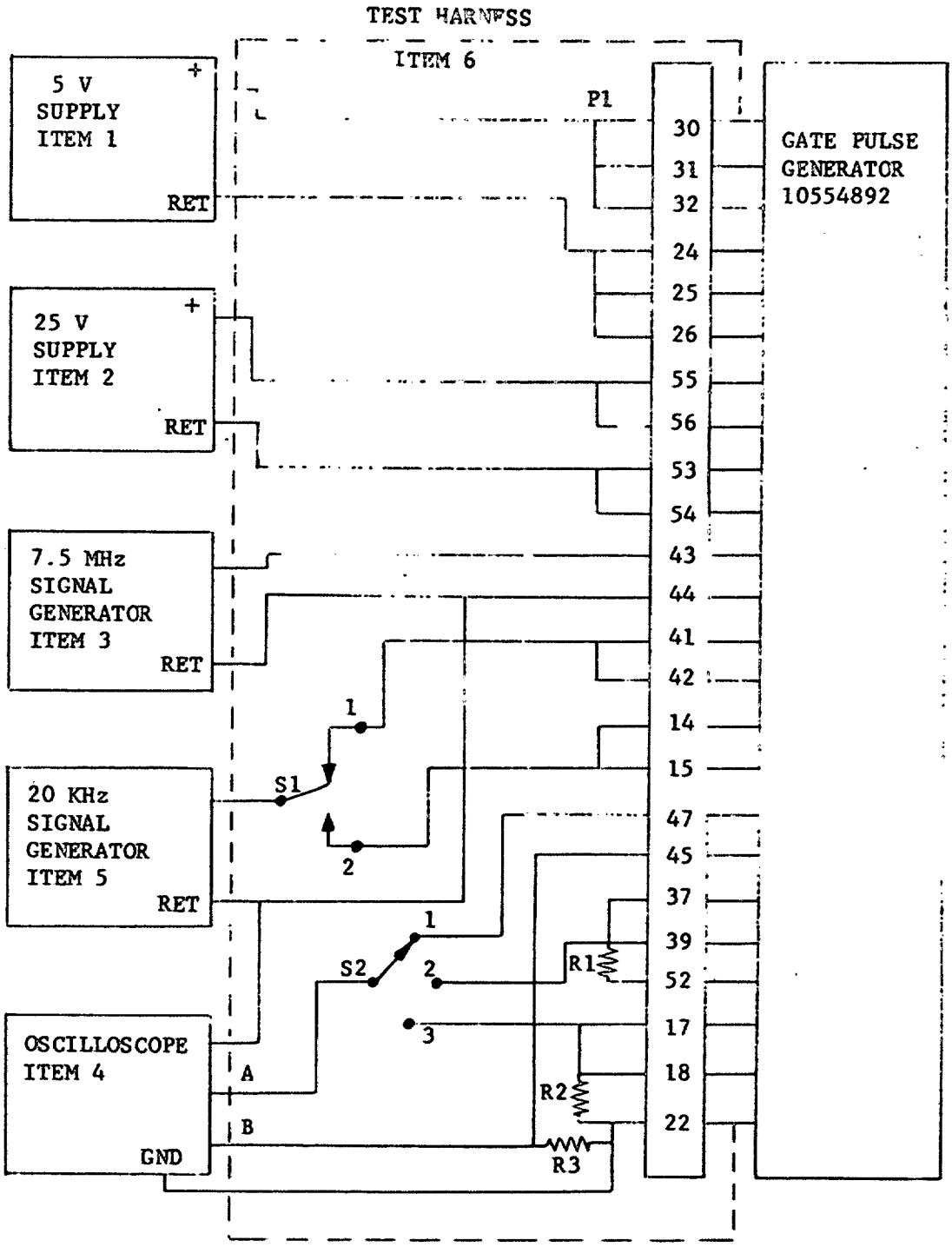


FIGURE I. TEST SET FOR GATE PULSE GENERATOR

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4.6 Test methods and procedures.

4.6.1 Environmental.- Environmental test shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

4.6.1.1 Vibration.- The test shall be performed as specified in Method 514, Procedure IX, Part I, of MIL-STD-810 except that Failure Criteria of Method 514 of MIL-STD-810 does not apply. After vibration, the gate pulse generator shall be tested as given in the sub-paragraphs of 4.6.2 and 4.6.3 for compliance with 3.3.1.

4.6.1.2 Shock.- This test shall be performed as specified in Procedure I of Method 516 of MIL-STD-810 except that shock pulses shall be 11.0 ± 1 ms and 15 g's. The G value of each impulse shall be within ±10 percent when measured with a 0.2 to 250 Hz filter and maximum G shall occur at approximately 5-1/2 milliseconds. After the shock impulses, the gate pulse generator shall be tested as specified in 4.6.2 and 4.6.3 for compliance with 3.3.2.

4.6.1.3 Altitude - non-operating and operating.- This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that in Step 2 the altitude shall be limited to 40,000 ft. (5.5 in Hg). Operation of Step 3 shall be the performance tests of 4.6.3.

4.6.1.4 Temperature.

4.6.1.4.1 High (operating and non-operating).- This test shall be performed as specified in Procedure I, Method 501, of MIL-STD-810. The temperature of Step 2 shall be +185°F. The temperature of Step 4 shall be +130°F. In Step 5 the gate pulse generator shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.4.2 Low (operating and non-operating).- This test shall be performed as specified in Procedure I, Method 502 of MIL-STD-810. The temperature of Step 2 shall be -85°F maintained for 24 hours. The temperature of Step 4 shall be +32°F. In Steps 5 and 7, the gate pulse generator shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.5 Humidity-non-operating.- This test shall be performed in accordance with Procedure I, Method 507 of MIL-STD-810 except that in Step 4 the relative humidity shall be maintained at 95%.

4.6.2 Electrical.- Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.1.

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4.6.3 Performance.- Adjust power supplies, item 1 and 2, to deliver outputs specified in Table I as ±5V and +25V, respectively. Turn off power supplies. Adjust signal generators, items 3 and 5, to delivery outputs specified in Table I as Cp and $\overline{PRF}$, respectively. Turn off signal generators. Connect test equipment as shown in Figure 1. Turn test harness switch S1 to position 1. Turn test harness switch S2 to position 1. Insert gate pulse generator into connector P1 of test harness. Turn on power supplies and signal generators. For measurement of rise time and pulse width see Figure 2A. Measurement of delay time, see Figure 2B.

a) Adjust oscilloscope and check that A channel signal meets requirements of "Cp out" of Table I.

b) Adjust oscilloscope and check that B channel signal meets requirements of "Cpd out" of Table I.

c) Turn switch S2 to position 2. Adjust oscilloscope and check that A channel signal meets requirements of "Output signal in response to $\overline{PRF}$" of Table I.

d) Turn switch S1 to position 2 and switch S2 to position 3. Check that A channel signal on oscilloscope meets requirements of "Output signal in response to $\overline{TARGET\ PULSE}$" of Table I.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging, packing, and marking shall be in accordance with Packaging Data Sheet F10554892. The level of protection shall be as specified in the contract.

6. NOTES

6.1 Intended use.- The gate pulse generator is used in Test Set, Subassembly MX-8048/TPM-22, part of Radar Test Set AN/TPM-22. The gate pulse generator amplifies two 20KHz signals, and provides a small delay in a 7.5 MHz signal, all of which are used elsewhere in Test Set, Subassembly MX-8048/TPM-22.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Applicable packaging data sheet (see 5.1).

(c) Selection of applicable level of preservation packaging and packing.

(d) Level of protection required.

(e) Applicable stock number.

(f) Block of serial numbers to identify each gate pulse generator.

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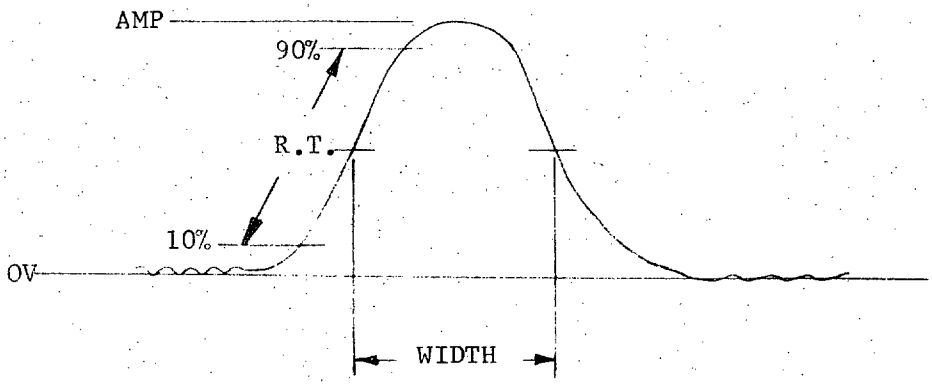


FIGURE 2A

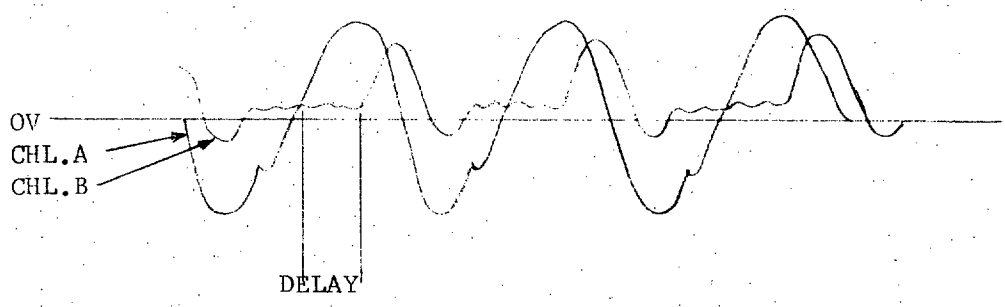


FIGURE 2B

Custodian

Army - MU

Preparing activity

Army - MU

Project No. 4931-A150

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-G-50349A(MI) Generator, Pulse: 1056302

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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